

CCNA Network Project Documentation

Cisco Packet Tracer Enterprise Network Simulation

Course: CCNA

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Platform: Cisco Packet Tracer

Project Type: Simple Network Simulation

1. Project Overview

This project is a practical Cisco Packet Tracer network simulation designed to apply the networking concepts covered during the CCNA course.

The network represents a small enterprise environment consisting of two separate IP networks. Each network contains a router connection, two switches, multiple end devices, a printer, and a dedicated DHCP server.

The project focuses on IPv4 addressing, subnetting, default gateways, DHCP configuration, switching, routing, and network troubleshooting.

The main objective of this project is to transform theoretical CCNA concepts into a functional network topology and verify communication between the different devices.

2. Project Objectives

The main objectives of this project are:

- Designing a functional enterprise network topology.
- Applying IPv4 addressing and subnetting.
- Dividing the `192.168.40.0/24` network into two subnets.
- Configuring router interfaces with appropriate IP addresses.
- Configuring default gateways for the different networks.
- Configuring dedicated DHCP servers.
- Assigning static IP addresses to network infrastructure devices and DHCP servers.
- Configuring PCs to obtain IP addresses automatically through DHCP.
- Connecting end devices through Cisco switches.
- Testing connectivity between network devices.

- Troubleshooting and resolving DHCP configuration problems.
 - Practicing Cisco IOS commands and network verification techniques.
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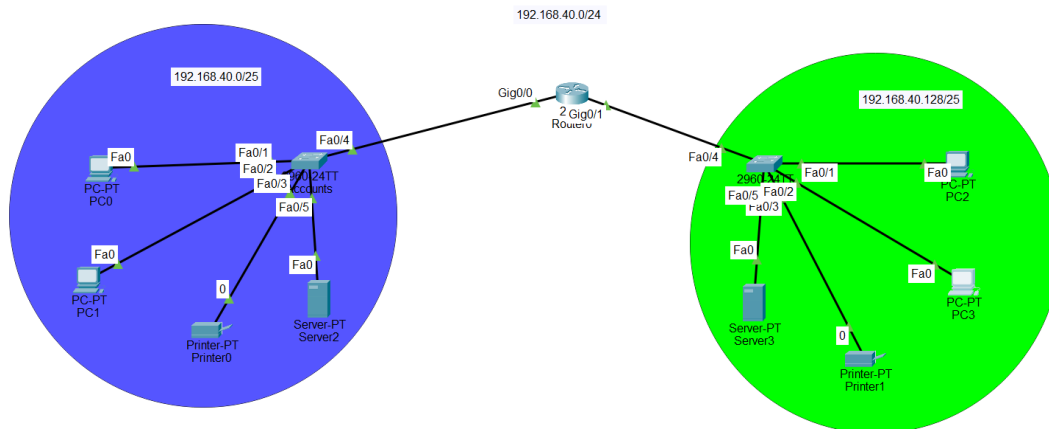
3. Network Topology

The network consists of two main network segments.

Each side contains:

- 1 Router
- 2 Switches
- 2 PCs per switch
- 1 Printer per switch
- 1 DHCP Server per switch/network

The general topology can be represented as:



Each switch provides Layer 2 connectivity between the connected end devices.

The routers provide Layer 3 connectivity between different IP networks.

4. IPv4 Addressing

The original network provided for the project was:

192.168.40.0/24

The /24 network was divided into two equal subnets using the subnet mask:

255.255.255.128

or:

/25

A /25 subnet provides:

- 128 total IP addresses
- 126 usable host addresses
- 1 network address
- 1 broadcast address

Therefore, the original /24 network was divided into the following two networks.

5. Subnetting Design

Network 1

Network Address:	192.168.40.0
Subnet Mask:	255.255.255.128
CIDR:	/25
Default Gateway:	192.168.40.1
First Usable IP:	192.168.40.1
Last Usable IP:	192.168.40.126
Broadcast Address:	192.168.40.127

The DHCP server was assigned:

IP Address:	192.168.40.3
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Subnet Mask: 255.255.255.128
Default Gateway: 192.168.40.1

The DHCP clients can therefore start receiving addresses from:

192.168.40.3

Network 2

Network Address: 192.168.40.128
Subnet Mask: 255.255.255.128
CIDR: /25
Default Gateway: 192.168.40.129
First Usable IP: 192.168.40.129
Last Usable IP: 192.168.40.254
Broadcast Address: 192.168.40.255

The DHCP server was assigned:

IP Address: 192.168.40.130
Subnet Mask: 255.255.255.128
Default Gateway: 192.168.40.129

The DHCP clients can therefore start receiving addresses from:

192.168.40.131

6. Addressing Table

Network	Network Address	Subnet Mask	Default Gateway	DHCP Server
Network 1	192.168.40.0/25	255.255.255.128	192.168.40.1	192.168.40.3
Network 2	192.168.40.128/25	255.255.255.128	192.168.40.129	192.168.40.131

7. Router Configuration

The router interfaces are configured with static IP addresses that act as the default gateways for the connected networks.

For Network 1:

IP Address: 192.168.40.1
Subnet Mask: 255.255.255.128

For Network 2:

IP Address: 192.168.40.129
Subnet Mask: 255.255.255.128

Example Cisco IOS configuration:

```
interface g0/0
 ip address 192.168.40.1 255.255.255.128
 no shutdown
```

For the second network:

```
interface g0/1
 ip address 192.168.40.129 255.255.255.128
 no shutdown
```

The exact interface numbers may vary depending on the router model used in Cisco Packet Tracer.

8. DHCP Server Configuration

Each network contains a dedicated DHCP server.

The DHCP servers were configured with static IP addresses because the DHCP server itself must have a stable and reachable IP address.

Network 1 DHCP Server

IP Address: 192.168.40.2
Subnet Mask: 255.255.255.128
Default Gateway: 192.168.40.1

DHCP Pool:

Network: 192.168.40.0/25
Default Gateway: 192.168.40.1
Subnet Mask: 255.255.255.128
Starting IP: 192.168.40.3

Network 2 DHCP Server

IP Address: 192.168.40.130
Subnet Mask: 255.255.255.128
Default Gateway: 192.168.40.129

DHCP Pool:

Network: 192.168.40.128/25
Default Gateway: 192.168.40.129
Subnet Mask: 255.255.255.128
Starting IP: 192.168.40.131

9. PC Configuration

The PCs were configured as DHCP clients.

Instead of manually assigning an IP address to every PC, DHCP automatically provides:

- IP Address
- Subnet Mask
- Default Gateway
- DNS information, if configured

The PCs were configured through:

Desktop
→ IP Configuration
→ DHCP

After receiving an IP address, the configuration can be verified using:

`ipconfig`

10. Printer Configuration

The network also includes printers connected to the switches.

Printers can be configured using static IP addresses so that they remain reachable at a predictable address.

A suitable design is to reserve specific addresses outside the dynamic DHCP range for infrastructure devices such as:

- Printers
- Servers
- Network management devices

This prevents IP address conflicts and makes network management easier.

11. Switching

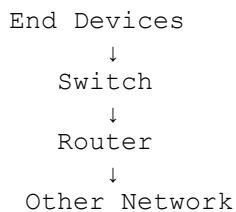
The switches provide Layer 2 connectivity between the connected devices.

Each switch connects:

- PCs
- Printer
- DHCP Server
- Router/network uplink

The switches forward Ethernet frames based on MAC addresses.

The project therefore demonstrates the basic relationship between:



12. Default Gateway

Each network uses the router interface as its default gateway.

For Network 1:

Default Gateway = 192.168.40.1

For Network 2:

Default Gateway = 192.168.40.129

The default gateway allows devices to communicate with destinations outside their local subnet.

For example, a PC in:

192.168.40.0/25

uses:

192.168.40.1

as its gateway when communicating with a device in:

192.168.40.128/25

13. Network Connectivity Testing

After completing the configuration, several tests were performed to verify the network.

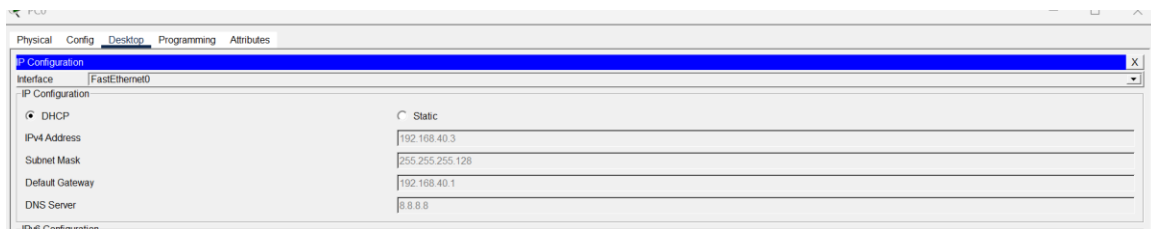
Test 1 – DHCP

The PCs were configured to use DHCP.

The expected result was that each PC would automatically receive an IP address from the appropriate DHCP server.

Example:

IP Address: 192.168.40.3
Subnet Mask: 255.255.255.128
Default Gateway: 192.168.40.1



Test 2 – Default Gateway

Each PC was tested against its default gateway.

Example:

ping 192.168.40.1

A successful reply confirms local connectivity between the PC and the router.

```
C:\>ping 192.168.40.1

Pinging 192.168.40.1 with 32 bytes of data:

Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time=15ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.40.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 15ms, Average = 3ms
```

Test 3 – Local Network Connectivity

Devices within the same subnet were tested using:

```
ping <destination-ip>
```

This verifies Layer 2 connectivity through the switch and IP connectivity within the subnet.

```
C:\>ping 192.168.40.4

Pinging 192.168.40.4 with 32 bytes of data:

Reply from 192.168.40.4: bytes=32 time<1ms TTL=128
Reply from 192.168.40.4: bytes=32 time<1ms TTL=128
Reply from 192.168.40.4: bytes=32 time<1ms TTL=128
Reply from 192.168.40.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.40.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Test 4 – Inter-Network Connectivity

Communication between devices in the two different subnets was tested.

For example:

```
Network 1
192.168.40.0/25
```

Router
↓
Network 2
192.168.40.128/25

Successful communication confirms that the router is correctly performing Layer 3 forwarding.

```
C:\>ping 192.168.40.131

Pinging 192.168.40.131 with 32 bytes of data:

Request timed out.
Reply from 192.168.40.131: bytes=32 time=10ms TTL=127
Reply from 192.168.40.131: bytes=32 time<1ms TTL=127
Reply from 192.168.40.131: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.40.131:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 3ms

C:\>tracert 192.168.40.131

Tracing route to 192.168.40.131 over a maximum of 30 hops:

  1  0 ms      0 ms      1 ms      192.168.40.1
  2  0 ms      0 ms      0 ms      192.168.40.131

Trace complete.
```

14. Cisco IOS Verification Commands

Several Cisco IOS commands can be used to verify the configuration.

Check interface status

```
show ip interface brief
```

This command displays:

- Interface names
 - IP addresses
 - Status
 - Protocol state
-

Check the running configuration

```
show running-config
```

This command displays the active configuration of the router.

Test connectivity

```
ping <destination-ip>
```

The ping command is used to test IP connectivity between devices.

15. Troubleshooting

During the implementation of the project, a DHCP problem occurred.

The PCs initially displayed:

```
DHCP Request Failed
```

This meant that the PCs were unable to obtain IP configuration automatically.

The problem was investigated by checking:

1. Physical connections.
2. Switch connectivity.
3. DHCP service status.
4. DHCP pool configuration.
5. Router gateway configuration.
6. DHCP server IP configuration.

The root cause was that the DHCP server itself did not have a proper static IP configuration.

16. DHCP Problem Resolution

The DHCP servers were configured with static IP addresses.

For Network 1:

192.168.40.2/25
Gateway: 192.168.40.1

For Network 2:

192.168.40.130/25
Gateway: 192.168.40.129

After configuring the DHCP servers correctly and enabling the DHCP service, the PCs successfully obtained IP addresses automatically.

This troubleshooting experience demonstrated an important practical networking concept:

A DHCP server must have a stable IP configuration and must be reachable by the clients it serves.

17. Technologies and Concepts Used

The project demonstrates the following CCNA concepts:

- IPv4 Addressing
 - IPv4 Subnetting
 - CIDR
 - /25 Subnetting
 - Network Address
 - Broadcast Address
 - Usable Host Addresses
 - Default Gateway
 - DHCP
 - Static IP Addressing
 - Dynamic IP Addressing
 - Ethernet Switching
 - MAC Address Learning
 - Layer 2 Connectivity
 - Layer 3 Routing
 - Cisco IOS
 - Network Troubleshooting
 - Connectivity Testing
 - Cisco Packet Tracer
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18. Skills Developed

Through this project, I developed practical experience in:

- Designing network topologies.
 - Calculating IPv4 subnets.
 - Selecting appropriate subnet masks.
 - Assigning IP addresses.
 - Configuring routers and switches.
 - Configuring DHCP services.
 - Understanding the role of default gateways.
 - Testing network connectivity.
 - Troubleshooting DHCP problems.
 - Using Cisco IOS verification commands.
 - Understanding the interaction between Layer 2 and Layer 3 devices.
-

19. Future Improvements

This project can be expanded into a more advanced enterprise network by implementing additional CCNA concepts such as:

VLANs

Create separate VLANs for different departments, such as:

```
VLAN 10 - IT
VLAN 20 - HR
VLAN 30 - Finance
VLAN 40 - Management
```

Trunking

Configure trunk links between switches to carry multiple VLANs.

Inter-VLAN Routing

Configure routing between VLANs using Router-on-a-Stick or Layer 3 switching.

OSPF

Replace static routing with dynamic routing using OSPF.

ACLs

Implement Access Control Lists to control communication between different networks.

STP

Configure Spanning Tree Protocol to prevent Layer 2 loops.

EtherChannel

Combine multiple physical links into a logical link to improve redundancy and bandwidth.

20. Project Outcome

The final network successfully demonstrates a functional enterprise-style topology using Cisco Packet Tracer.

The project includes two /25 IPv4 networks created from the original 192.168.40.0/24 address space.

Each network contains switches, PCs, printers, and a dedicated DHCP server. The routers provide the required Layer 3 connectivity, while DHCP provides automatic IP configuration to the client PCs.

The project also provided practical troubleshooting experience by identifying and resolving a DHCP configuration issue.

21. Conclusion

This project represents my first practical networking project while studying for the CCNA certification.

It allowed me to move beyond theoretical concepts and apply networking knowledge in a simulated enterprise environment.

The project provided hands-on experience with IPv4 subnetting, DHCP, switching, routing, default gateways, IP addressing, network verification, and troubleshooting.

Most importantly, the project demonstrated that successful networking is not only about configuring devices, but also about understanding how different network components interact and how to troubleshoot problems when the expected result is not achieved.

This project serves as a foundation for more advanced CCNA projects and future enterprise networking implementations.